

DiaTool-DPO: Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

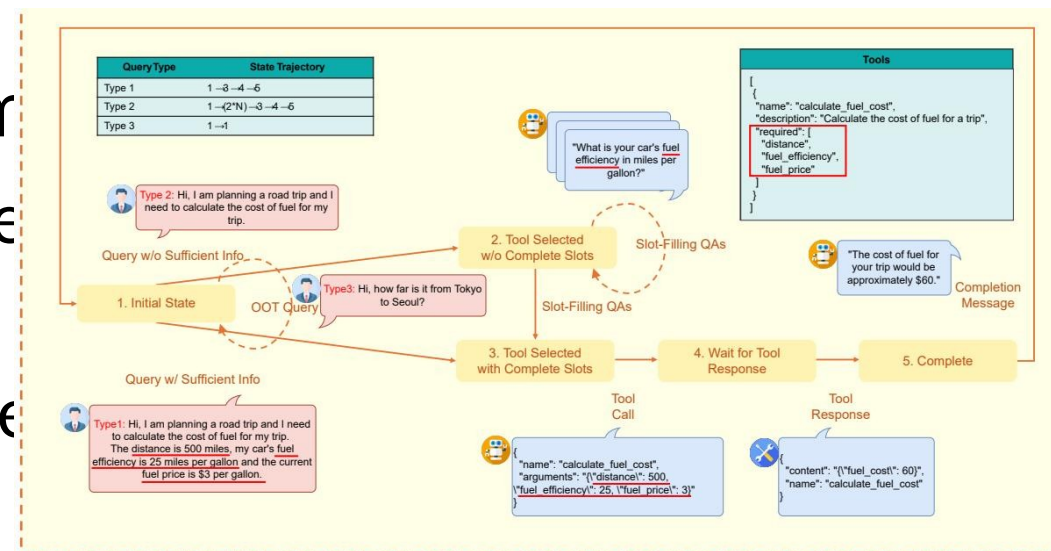
A novel DPO-based method that models TA-LLM interactions as an MDP with 5 dialogue states and 3 query types, automatically constructing preference pairs without human labeling.

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

- Core Problem: TA-LLMs trained only with Supervised Fine-Tuning (SFT) struggle with conversational flow control.
- Three primary failure modes in SFT-only models:
 - 1. Failing to ask follow-up questions for missing information (incomplete queries).
 - 2. Making hallucinated tool invocations when slot values are ungrounded.
 - 3. Failing to reject inappropriate or out of scope

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

- 5 Internal States of the MDP Form
 - State 1: Initial State (No history, reject scope here)
 - State 2: Tool Selected w/o Complete slot filling QA occurs here)
 - State 3: Tool Selected w/ Complete Slots
 - State 4: Wait for Tool Response
 - State 5: Complete (Results received, response generated)



Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description & Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Contains all required args. No slot-filling needed. Example: "Book a flight from NY to LA on Oct 5th."
Type 2	$1 \rightarrow (2 * N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Lacks some args. Repeats QA N times to gather info. Example: "Book a flight to LA." (System asks for origin/date)
Type 3	$1 \rightarrow 1$	Functionality unavailable. Out-of-scope, rejected. Example: "Make me a sandwich." (System refuses politely)

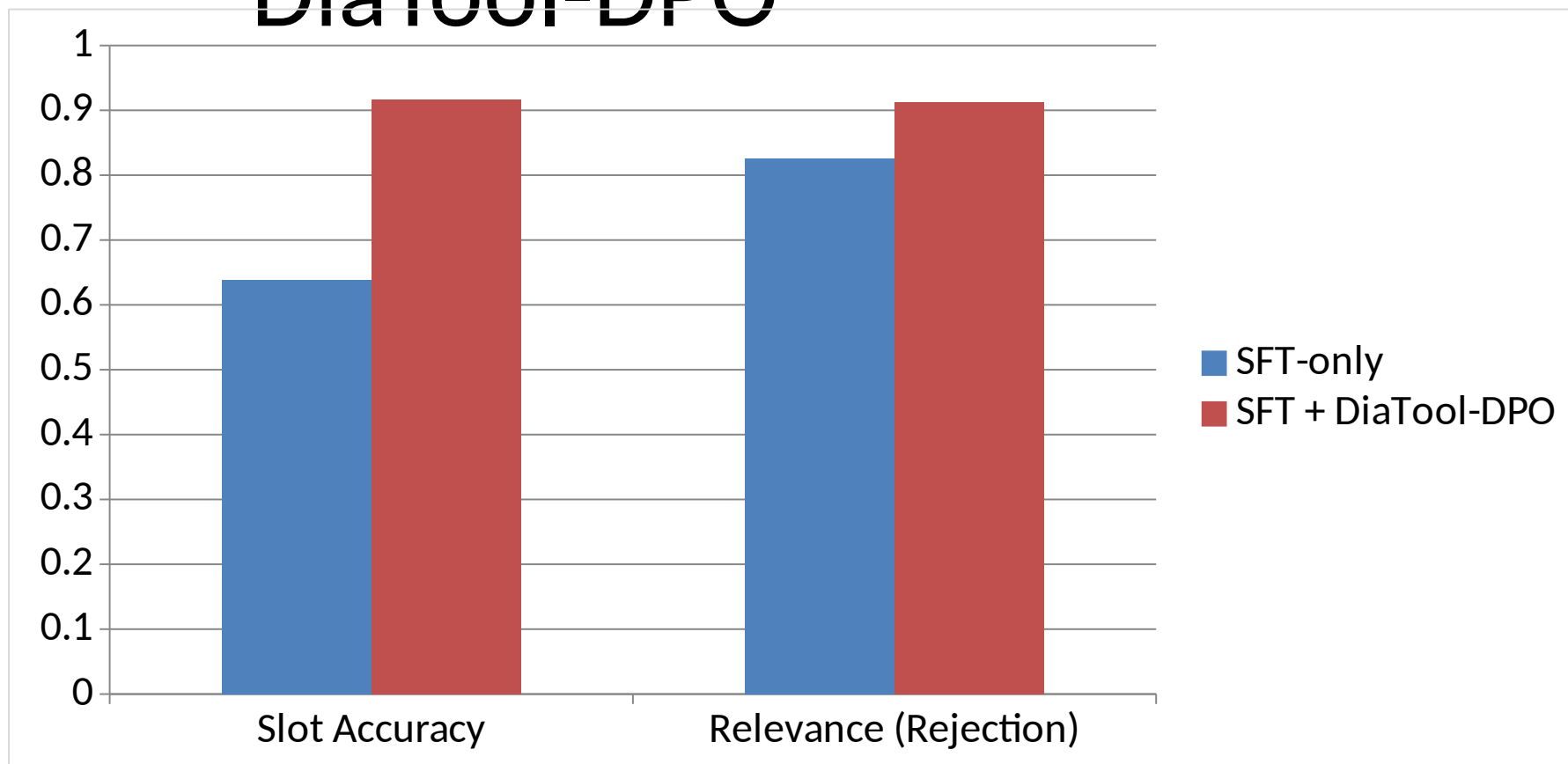
Automatic Paired Trajectory Construction Without Human Labels

Query Type	Chosen Trajectory	Rejected Trajectory	Learning Lesson	Count
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089 (Easy)
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	3,092 (Hard)
Type 1	1→3→4→5	1→1	Tool call accept	2,652 (Mix)
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089 (Easy)
Type 2	1→(2*N)→3→4→5	1→3→4→5 (or M)	Prevent slot hallucination	3,092 (Hard)
Type 2	—	1→1	Tool call accept	2,651 (Mix)
Type 3	1→1	1→3→4 (or N)	Tool call reject	1,129 (Hard)

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Adding DiaTool-DPO provides massive gains in conversational control (Slot-filling and Tool Rejection), pushing LLaMA-3-8B significantly closer to GPT-4o-level capability.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

- MDP Formulation Enables Systematic Preference Pairing:
 - Defining 5 internal states and 3 query types generates chosen/rejected trajectory pairs systematically.
- No Human Labeling Required:
 - Automatic dataset construction eliminates the costly bottleneck of human preference annotation.
- DPO Complements SET.